DBMS PROJECT • REVIEW 3

Cinema Screening and Ticket Booking System

Python + MySQL application design and validation

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Project Objectives

01 Connect Python application securely to MySQL

02 Implement CRUD for core cinema entities

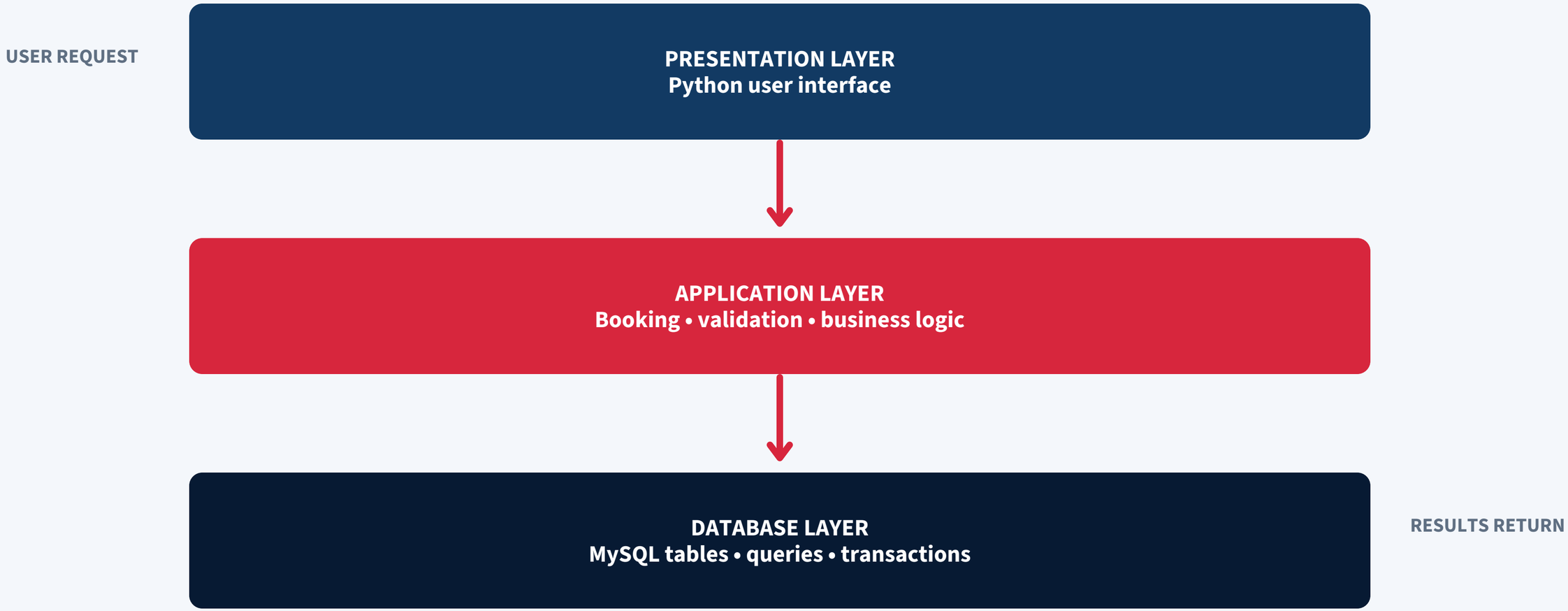
03 Manage schedules and live seat availability

04 Book seats and issue customer tickets

05 Process concessions, payments and refunds

06 Generate management-ready operational reports

System Architecture



Application Modules

Cinema & screens

Maintain halls, screens and capacities

Movies & shows

Schedule timings; check conflicts

Customers

Profiles and booking history

Seats & booking

Availability and allocation updates

Tickets & concessions

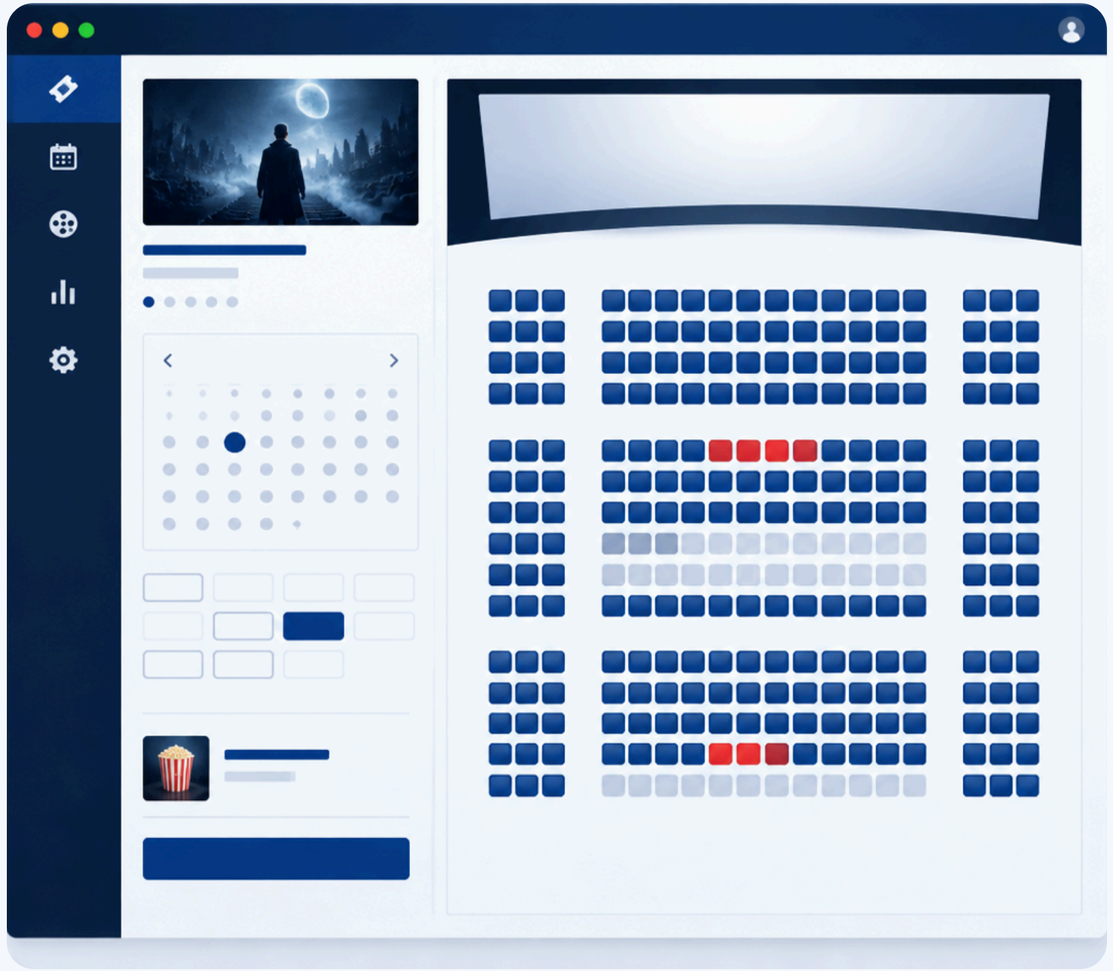
Orders, payments and ticket issue

Controls & reports

Cancellations, refunds and analytics

Show Scheduling & Seat Availability

- Select a movie, screen and show date
- Define times; reject overlapping screenings
- Refresh seat state before confirmation
- Lock allocation to prevent double booking
- Illustrative UI — validation logic implemented



ILLUSTRATIVE APPLICATION MOCKUP

Booking Workflow



Transactional boundary

Seat allocation and payment commit together. Any failure triggers rollback; no partial booking is retained.

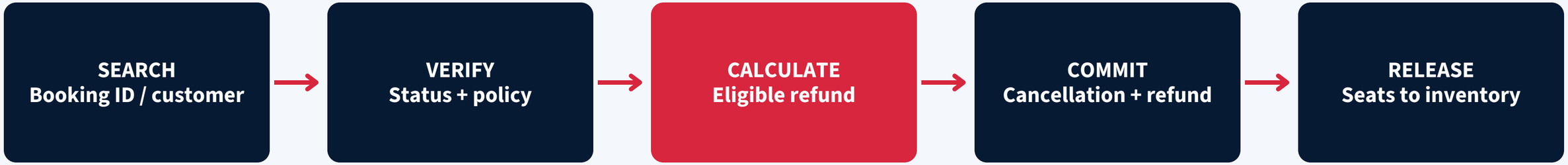
Concessions & Payments



ILLUSTRATIVE APPLICATION MOCKUP

- Select snacks and beverages
- Enter valid quantities; calculate totals
- Validate amount and payment status
- Persist order and payment details
- Report concession sales by item and date
- Illustrative UI — payment gateway Not Tested

Cancellation & Refunds



Policy gate

Reject invalid or ineligible requests before any database update.

Atomic recovery

Refund record and seat release are committed together, or rolled back.

Database Connectivity & CRUD

- Create, Read, Update and Delete core records
- Use parameterised queries against SQL injection
- Open, use and close MySQL connections safely
- Commit successful writes; roll back exceptions
- Credentials should be environment-configured

PYTHON / MYSQL

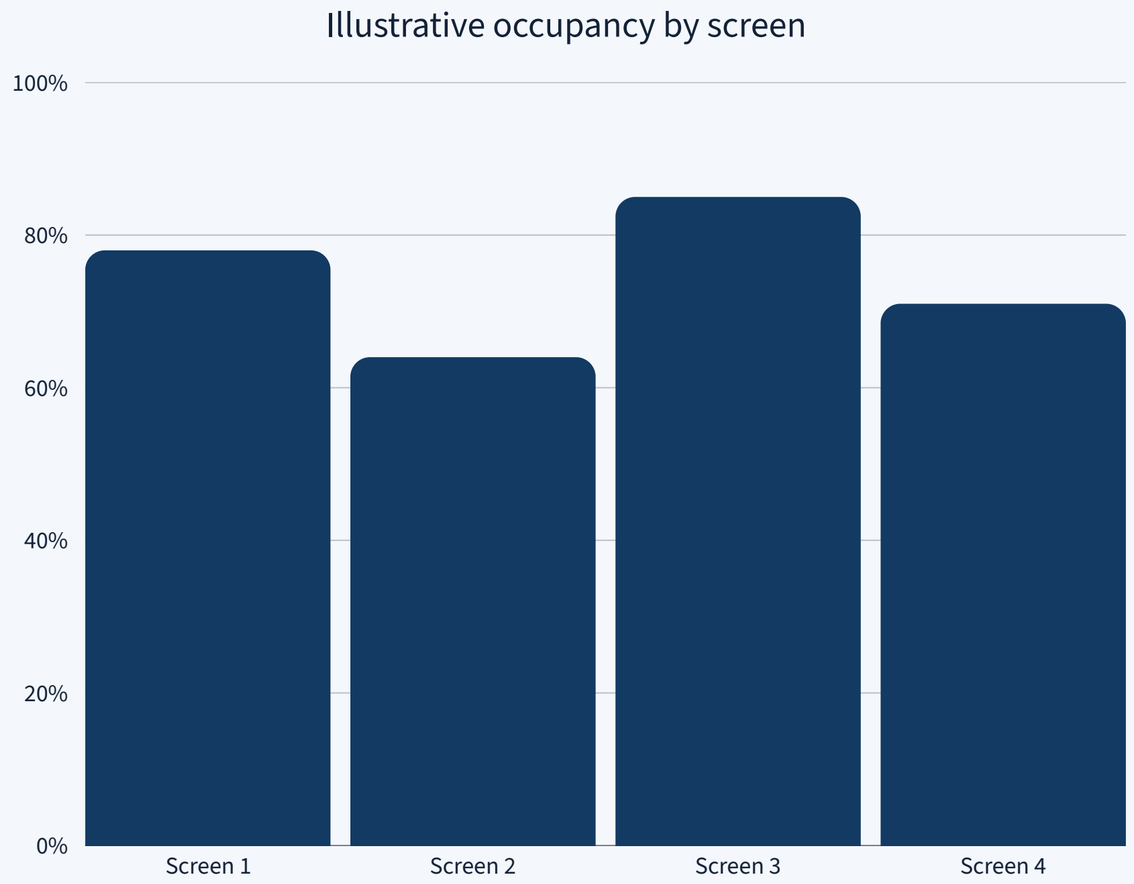
```
import mysql.connector

conn = mysql.connector.connect(**db_cfg)
cur = conn.cursor(dictionary=True)

cur.execute(
  "SELECT * FROM shows WHERE movie_id=%s",
  (movie_id,)
)
rows = cur.fetchall()
conn.close()
```

Reports & Search Features

- Occupancy by show and screen
- Movie-wise ticket sales summary
- Screen utilisation statistics
- Concession revenue by item and date
- Cancellation and refund summaries
- Queries use JOIN, GROUP BY, aggregates and views



Testing & Error Handling

TEST CASE	EXPECTED RESULT	STATUS
Duplicate seat booking	Rejected	PASS
Zero concession quantity	Rejected	PASS
Overlapping screenings	Rejected	PASS
Invalid booking ID	Error message	PASS
Refund above eligibility	Rejected	PASS
Failed booking transaction	Rollback	NOT TESTED

Final Demonstration

- Add or search movies; schedule screenings
- Check availability and book selected seats
- Record concessions and payment details
- Cancel eligible bookings and process refunds
- Display operational reports and analytics
- Deliver: source, SQL scripts, evidence, report and viva prep

THANK YOU

Reliable seat management.
Organised booking.
Accurate reporting.

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